

Monte Carlo Tests

Part II ■ Multiple Testing and Simulation-Based Inference

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Pivotal statistics

The classical linear regression (CLR) model

$$y = X\beta + \varepsilon, \quad \varepsilon \mid X \sim N_n(0, \sigma^2 I_n),$$

with X fixed, full column rank k .

Residual maker.

$$M = I_n - X(X'X)^{-1}X'$$

Symmetric, idempotent, rank $n - k$.

Two key facts.

- $MX = 0$, so $\hat{\varepsilon} = My = M\varepsilon$ (residuals see only the noise)
- $\hat{\sigma}^2 = \frac{\varepsilon' M \varepsilon}{n - k}$

The t -statistic, written out

Under $H_0 : \beta_j = \beta_j^0$:

$$t_j = \frac{e_j'(X'X)^{-1}X'\varepsilon / \sqrt{[(X'X)^{-1}]_{jj}}}{\sqrt{\varepsilon'M\varepsilon / (n - k)}}.$$

- Numerator depends on ε through $X'\varepsilon$ (the column space of X)
- Denominator depends on ε through $M\varepsilon$ (the orthogonal complement)
- Under normality, these two pieces are **independent**.

The σ cancels: pivotality

Write $\varepsilon = \sigma u$ with $u \sim N_n(0, I_n)$:

$$t_j = \frac{e'_j(X'X)^{-1}X' \cancel{\sigma} u / \sqrt{[(X'X)^{-1}]_{jj}}}{\sqrt{\cancel{\sigma}^2 u' M_X u / (n - k)}} = \frac{e'_j(X'X)^{-1}X' u / \sqrt{[(X'X)^{-1}]_{jj}}}{\sqrt{u' M_X u / (n - k)}},$$

where $M_X = I_n - X(X'X)^{-1}X'$.

What's left is a function of $u \sim N_n(0, I_n)$ and the fixed matrix X alone.

No β . No σ . The null distribution depends only on X and $n - k$.

A statistic whose null distribution does not depend on unknown parameters is pivotal.

The F -statistic, written out

Under $H_0 : R\beta = r$, with R a $q \times k$ matrix of full row rank:

$$F = \frac{(R\hat{\beta} - r)' [R(X'X)^{-1}R']^{-1} (R\hat{\beta} - r) / q}{\varepsilon' M \varepsilon / (n - k)}.$$

- Numerator: $R\hat{\beta} - r = R(X'X)^{-1}X'\varepsilon$ under H_0 (depends on $X'\varepsilon$)
- Denominator: $\hat{\sigma}^2 = \varepsilon' M \varepsilon / (n - k)$ (depends on $M\varepsilon$)
- Same geometric split as the t -statistic.

The F -statistic is pivotal

Substitute $\varepsilon = \sigma u$ with $u \sim N_n(0, I_n)$:

$$F = \frac{\cancel{\sigma^2} (R(X'X)^{-1}X'u)' [R(X'X)^{-1}R']^{-1} R(X'X)^{-1}X'u / q}{\cancel{\sigma^2} u' M_X u / (n - k)}.$$

What's left is a function of $u \sim N_n(0, I_n)$ and the fixed matrix X .

No β . No σ .

Same story as t : studentization cancels the nuisance parameter.
The F -statistic is pivotal.

When the textbook runs out

t and F are pivotal in the CLR model.

They also have textbook null distributions: t_{n-k} and $F_{q, n-k}$.

Pivotality is the general property. The closed-form distribution is the lucky accident.

Many useful statistics are pivotal under the CLR model but their exact finite-sample null distribution is hard or impossible to evaluate in closed form:

- $\max_j |t_j|$ over several coefficients: requires a multivariate t integral
- Jarque–Bera, Breusch–Pagan, Engle's ARCH: asymptotic χ^2 , not exact
- Any statistic you can write as a function of $u = \varepsilon/\sigma$ and X

When the reference distribution is hard to evaluate: *simulate it*.

The Monte Carlo recipe

Let T be a continuous pivotal statistic under H_0 , with large values taken as evidence against H_0 (right-tailed test).

Recipe.

1. Compute the observed statistic from the data, call it T_B .
2. Draw $u^{(1)}, \dots, u^{(B-1)} \sim N_n(0, I_n)$ and compute simulated statistics $T_1, \dots, T_{B-1}$ under H_0 .
3. Rank T_B among $\{T_1, \dots, T_{B-1}, T_B\}$ and let $\hat{R}(T_B)$ be its rank. The Monte Carlo p -value is

$$\hat{p}(T_B) = \frac{B - \hat{R}(T_B) + 1}{B}.$$

4. Reject H_0 at level α if $\hat{p}(T_B) \leq \alpha$.

Equivalently: reject if $\hat{R}(T_B) \geq B(1 - \alpha) + 1$.

For $\alpha = 0.05$ and $B = 100$: reject if T_B ranks at least 96 out of 100.

Why it's exact

Three steps.

T pivotal under $H_0 \implies T_1, \dots, T_B$ are **exchangeable** under H_0

Exchangeability $\implies \hat{R}(T_B)$ is **uniform** on $\{1, 2, \dots, B\}$

Uniformity $\implies P_{H_0} \left[\hat{R}(T_B) \geq B(1 - \alpha) + 1 \right] = \alpha$

The level is **exact**: not asymptotic in n , not asymptotic in B .

Only requires αB to be an integer.

For $\alpha = 0.05$, one can use $B = 20, 100, 200, 1000, \dots$

B is a power knob, not a validity knob.

Max- t and the joint null distribution

The max- t statistic as a multiple test

Test m slope hypotheses $H_j : \beta_j = 0, j = 1, \dots, m$, simultaneously.

The test statistic.

$$T_{\max} = \max_{j=1, \dots, m} |t_j|$$

Reject H_j if $|t_j| \geq u_{1-\alpha}$, where $u_{1-\alpha}$ solves

$$P(T_{\max} \leq u_{1-\alpha} \mid H_1 \cap \dots \cap H_m) = 1 - \alpha$$

The obstacle. Under the global null, $(t_1, \dots, t_m)$ follows an m -dimensional multivariate t with $\nu = n - k$ degrees of freedom and correlation matrix R from $(X'X)^{-1}$. Evaluating $u_{1-\alpha}$ requires integrating over this joint distribution.

Adjusted p -value for H_j .

$$\tilde{p}_j = P(T_{\max} \geq |t_j^{\text{obs}}| \mid H_1 \cap \dots \cap H_m)$$

The integral that multcomp evaluates

Hothorn, Bretz & Westfall (2008), eq. (5).

The probability that all m statistics fall inside the symmetric acceptance box:

$$g(\hat{R}, t) = \underbrace{\int_{-t}^t \cdots \int_{-t}^t}_{m \text{ integrals}} \phi_m(x_1, \dots, x_m; \hat{R}, \nu) dx_1 \cdots dx_m,$$

where $\phi_m(\cdot; \hat{R}, \nu)$ is the density of $t_m(\nu, \hat{R})$, $\nu = n - k$, and $\hat{R}$ is estimated from $(X'X)^{-1}$.

Adjusted p -value for H_j :

$$\tilde{p}_j = 1 - g(\hat{R}, |t_j^{\text{obs}}|).$$

What multcomp does.

- Estimate $\hat{R}$ from the fitted model.
- Evaluate $g(\hat{R}, |t_j|)$ numerically.
- Return $1 - g(\hat{R}, |t_j|)$ as the adjusted p -value.

For $m = 9$ (body fat): a 9-dimensional integral.

All m statistics are functions of M and u only

Under $H_1 \cap \dots \cap H_m$, substitute $\varepsilon = \sigma u$, $u \sim N_n(0, I_n)$. Each t_j reduces to:

$$t_j = \frac{e'_j(X'X)^{-1}X'u / \sqrt{[(X'X)^{-1}]_{jj}}}{\sqrt{u'M_X u / (n - k)}}$$

Let $C = (X'X)^{-1}$, let S select the m tested coefficients, and define $D = \text{diag}(SCS')$, $A = D^{-1/2}SCX'$.

Stacking all m statistics:

$$\begin{pmatrix} t_1 \\ \vdots \\ t_m \end{pmatrix} = \frac{Au}{\sqrt{u'M_X u / (n - k)}}.$$

Note: $u \sim N_n(0, I_n)$ is the only source of randomness; A and M_X are fixed, determined by X .

No β . No σ . $(t_1, \dots, t_m)$ is a deterministic function of u and X .

**Under the complete null, $(t_1, \dots, t_m)$ is jointly pivotal:
the entire vector is a deterministic function of $u \sim N_n(0, I_n)$ and X alone.
So is any function of it, including $T_{\max} = \max_j |t_j|$.**

The MC max- t procedure

Pivotality makes exact simulation possible. Apply the MC recipe to $T_{\max}$:

1. Compute $t_1^{\text{obs}}, \dots, t_m^{\text{obs}}$ from the data and set $T_{\max, B} = \max_j |t_j^{\text{obs}}|$.
2. For $b = 1, \dots, B-1$: draw $u^{(b)} \sim N_n(0, I_n)$ and compute

$$\begin{pmatrix} t_1^{(b)} \\ \vdots \\ t_m^{(b)} \end{pmatrix} = \frac{A u^{(b)}}{\sqrt{u^{(b)'} M_X u^{(b)} / (n - k)}}, \quad T_{\max}^{(b)} = \max_j |t_j^{(b)}|.$$

3. Rank $T_{\max, B}$ among $\{T_{\max}^{(1)}, \dots, T_{\max}^{(B-1)}, T_{\max, B}\}$:

$$\hat{p}(T_{\max, B}) = \frac{B - \hat{R}(T_{\max, B}) + 1}{B}.$$

**Same joint null distribution as `multcomp`, simulated rather than integrated.
Correlation structure is captured automatically. No formula to look up.**

Body fat: MC max- t versus multcomp

$m = 9$ slopes, $n = 71$, $\nu = 61$. $B = 1000$ draws.

	unadj.	multcomp (max- t)	MC max- t
age	0.538	0.996	0.996
waistcirc	0.003	0.021*	0.021*
hipcirc	<0.001	<0.001***	<0.001***
elbowbreadth	0.688	1.000	1.000
kneebreadth	0.018*	0.132	0.146
anthro3a	0.274	0.895	0.891
anthro3b	0.086	0.478	0.484
anthro3c	0.853	1.000	1.000
anthro4	0.315	0.930	0.922

The two adjusted columns agree to two or three decimal places. kneebreadth does not survive under either procedure. waistcirc and hipcirc survive both.

Simulation and numerical integration estimate the same quantity because they target the same joint null distribution.

What the MC approach buys

`multcomp` (numerical integration).

- Evaluates the m -dimensional multivariate t integral numerically.
- Exact to numerical precision.
- Requires $(t_1, \dots, t_m) \sim t_m(\nu, R)$ with R from $(X'X)^{-1}$.
- Fast: one integral call per hypothesis.

MC max- t .

- Draws $u \sim N_n(0, I_n)$ and computes $(t_1, \dots, t_m) = Au / \sqrt{u' M_X u / (n - k)}$.
- No integral. No distributional formula.
- Correctness guaranteed by pivotality and exchangeability.
- Generalises immediately to any statistic that is a function of u and X .
- Cost: B evaluations of the t -vector formula.

**Both methods exploit the same pivotality.
The MC approach makes that pivotality visible and generalises it.**

Why single-step leaves power on the table

Single-step ranks every H_j against the same reference distribution:

$$T_{\max} = \max_{j=1,\dots,m} |T_j|.$$

Step-down uses the closure idea. After rejecting $H_{(1)}$, the remaining hypotheses are compared with a reduced maximum, such as $\max_{j \geq 2} |T_j|$, which has a less conservative reference distribution.

Reduced-set distributions.

- Subset pivotality gives exactness at every step.
- Correlation among the remaining statistics is preserved.
- The same simulated draws $\{u^{(b)}\}$ are reused; only the index set changes.

Step-down uses successively smaller maxima.
Subset pivotality makes the reduced-set reference distributions valid.

Why step-down is valid: subset pivotality

The key condition is **subset pivotality**. For any subset $J \subseteq \{1, \dots, m\}$, under

$$H_J : \beta_j = 0, \quad j \in J,$$

the joint distribution of $(T_j : j \in J)$ does not depend on the nuisance coefficients β_ℓ , $\ell \notin J$.

In the classical Gaussian linear model, write $\varepsilon = \sigma u$, with $u \sim N_n(0, I_n)$. Then

$$\hat{\beta} - \beta = \sigma(X'X)^{-1}X'u, \quad \hat{\sigma}^2 = \sigma^2 \frac{u' M_X u}{n - k}.$$

Hence, under H_J ,

$$T_J = D_J^{-1/2} \frac{S_J(X'X)^{-1}X'u}{\sqrt{u' M_X u / (n - k)}}.$$

This depends only on u and X , not on β_{-J} or σ .

**Subset pivotality makes the reduced-set max- t distributions valid.
They can therefore be simulated under the complete null.**

Step-down max- t : the adjusted p -values

Sort the observed statistics:

$$|t_{(1)}^{\text{obs}}| \geq \cdots \geq |t_{(m)}^{\text{obs}}|.$$

Use this same observed ordering for the simulated vectors.

$$\tilde{p}_{\text{SD}}(|t_{(1)}^{\text{obs}}|) = P\left(\max_{j=1, \dots, m} |T_{(j)}| \geq |t_{(1)}^{\text{obs}}| \mid H_0\right),$$

$$\tilde{p}_{\text{SD}}(|t_{(2)}^{\text{obs}}|) = \max\left\{\tilde{p}_{\text{SD}}(|t_{(1)}^{\text{obs}}|), P\left(\max_{j \geq 2} |T_{(j)}| \geq |t_{(2)}^{\text{obs}}| \mid H_0\right)\right\},$$

$$\vdots$$

$$\tilde{p}_{\text{SD}}(|t_{(k)}^{\text{obs}}|) = \max\left\{\tilde{p}_{\text{SD}}(|t_{(k-1)}^{\text{obs}}|), P\left(\max_{j \geq k} |T_{(j)}| \geq |t_{(k)}^{\text{obs}}| \mid H_0\right)\right\},$$

$$\vdots$$

$$\tilde{p}_{\text{SD}}(|t_{(m)}^{\text{obs}}|) = \max\left\{\tilde{p}_{\text{SD}}(|t_{(m-1)}^{\text{obs}}|), P(|T_{(m)}| \geq |t_{(m)}^{\text{obs}}| \mid H_0)\right\}.$$

- Step 1 is the single-step adjusted p -value.
- Steps $k \geq 2$ use reduced maxima $\max_{j \geq k} |T_{(j)}|$.
- The outer max enforces monotonicity.

The MC step-down recipe

Reuse the same $B - 1$ simulated draws. Sort the observed statistics:

$$|t_{(1)}^{\text{obs}}| \geq \cdots \geq |t_{(m)}^{\text{obs}}|,$$

and use this same hypothesis ordering for every simulated vector.

1. For each draw b and step k , compute

$$T_{\max,k}^{(b)} = \max_{j \geq k} |t_{(j)}^{(b)}|.$$

2. At step k , rank $|t_{(k)}^{\text{obs}}|$ against $\{T_{\max,k}^{(b)}\}_{b=1}^{B-1}$:

$$p_k^{\text{raw}} = \frac{B - \hat{R}_k(|t_{(k)}^{\text{obs}}|) + 1}{B}.$$

3. Enforce monotonicity:

$$\tilde{p}_{\text{SD}}(|t_{(1)}^{\text{obs}}|) = p_1^{\text{raw}}, \quad \tilde{p}_{\text{SD}}(|t_{(k)}^{\text{obs}}|) = \max\left\{\tilde{p}_{\text{SD}}(|t_{(k-1)}^{\text{obs}}|), p_k^{\text{raw}}\right\}.$$

4. Recover original order: assign each H_j its step- k adjusted p -value.

**MC step-down reuses the same draws, but changes the max at each step.
Subset pivotality ensures exactness of the reduced-set reference distributions.**

Demo 1: MC max- t on the body fat data

```
demo_bodyfat_maxt.R
```

- Precompute $A = D^{-1/2}(X'X)^{-1}X'$ and $M_X = I_n - X(X'X)^{-1}X'$ from the design matrix.
- Draw $B - 1 = 999$ vectors $u \sim N_n(0, I_n)$.
- Build single-step and step-down adjusted p -values.
- Compare with `multcomp` reference.

When pivotality fails

Pivotality is the exception, not the rule

Everything so far worked because $(t_1, \dots, t_m)$ is pivotal: a deterministic function of u and X , simulable exactly.

Many test batteries have no usable pivot.

In a battery of nonlinearity tests, each statistic exploits different features of the estimated residuals. Their joint null distribution depends on the DGP in ways that cannot be removed by studentization.

Without a pivot, two options remain.

- **Distributional inequalities.** Bonferroni or Holm: simple, always valid, but conservative.
- **Bootstrap approximation.** Resample from the fitted model: accounts for correlation, but only asymptotically valid.

Pivotality is the price of exactness.

When it fails, we bound the error or approximate it: we cannot eliminate it.

Testing for nonlinearity: a multiplicity problem

Running $k = 7$ tests and rejecting if *any* is significant is methodologically appealing. But the asymptotic FWER of the naive battery is not α_0 .

Asymptotic FWER with $k = 7$ tests:

Nominal α_0	Under independence $1 - (1 - \alpha_0)^7$	Psaradakis simulation
5%	30.2%	27.2%
10%	52.2%	45.8%

**The naive battery has asymptotic FWER $\gg \alpha_0$.
Declaring significance when any one test rejects is not a test at level α_0 .**

The model and the test battery

The null model.

$$x_t = \mu + \sum_{j=1}^m \varphi_j x_{t-j} + \varepsilon_t, \quad \{\varepsilon_t\} \stackrel{i.i.d.}{\sim} (0, \sigma^2),$$

m selected by AICc (Hurvich & Tsai, 1989) over $m \in \{1, \dots, 8\}$.

Seven tests, each targeting a distinct departure from linearity.

#	Test	Targeted alternative
1	Keenan	Quadratic terms in lagged x_t
2	Tsay	Threshold and interaction structure
3	RESET (modified, powers ≤ 4)	General polynomial misspecification
4	McLeod-Li (20 lags)	ARCH and nonlinear variance effects
5	BDS (embedding dim. = 2)	Any departure from i.i.d. residuals
6	V23 neural-network test	Logistic activation nonlinearity
7	NEURAL2 neural-network test	Random-projection activations

The tests share the same $\hat{\varepsilon}_t$: their p-values are **correlated** and their joint null distribution is intractable.

The tests are complementary, not redundant.
Their joint null distribution has no pivot: it cannot be simulated exactly.

Multiplicity-adjusted p-values

The goal. Find $\tilde{p}_1, \dots, \tilde{p}_k$ such that $P(V \geq 1) \leq \alpha$. Reject H_i when $\tilde{p}_i \leq \alpha$.

Two closed-form adjustments. Sort $p_{(1)} \leq \dots \leq p_{(k)}$.

Method	Adjusted p-value	Key property
Bonferroni	$\min(k p_i, 1)$	Union bound; no dependence assumption
Holm	$q_{(i)} = \min\{1, \max[(k - i + 1) p_{(i)}, q_{(i-1)}]\}$	Step-down; uniformly sharper

Bootstrap adjustment. Resample $\{\hat{\varepsilon}_t\}$, propagate through $\text{AR}(m)$, refit, recompute all k p-values:

$$\tilde{p}_i^{(N)} = \frac{1}{N} \sum_{n=1}^N \mathbf{1} \left[\min_{1 \leq j \leq k} p_{j,n}^* \leq p_i \right].$$

All three are asymptotically valid: marginal p-values are not exact here.
Bonferroni and Holm need only valid marginals;
the bootstrap also accounts for correlation.

Demo 2: Nonlinearity testing on US GDP

`demo_psaradakis.R`

- Seven nonlinearity tests on US GDP growth, AR(2) null.
- Bonferroni, Holm, and residual bootstrap adjustments.
- Only McLeod-Li has any signal: does it survive correction?
- Effect of GDP data revisions on the conclusion.